

What is an I/O Module?

An I/O module is a hardware component serving as an interface between a computer's central processing unit (CPU) and input/output devices. It connects the processor's system bus and one or more peripheral devices (input/output) devices.

A system bus is a network of electrical wires and cables connecting major components of a computer, such as the CPU, memory, and I/O devices.

The I/O module is responsible for handling the communication between the CPU and input and output devices. This includes managing data transfer, controlling power loads, and handling machine instructions. It enables system integrators to connect to different and more number of devices.

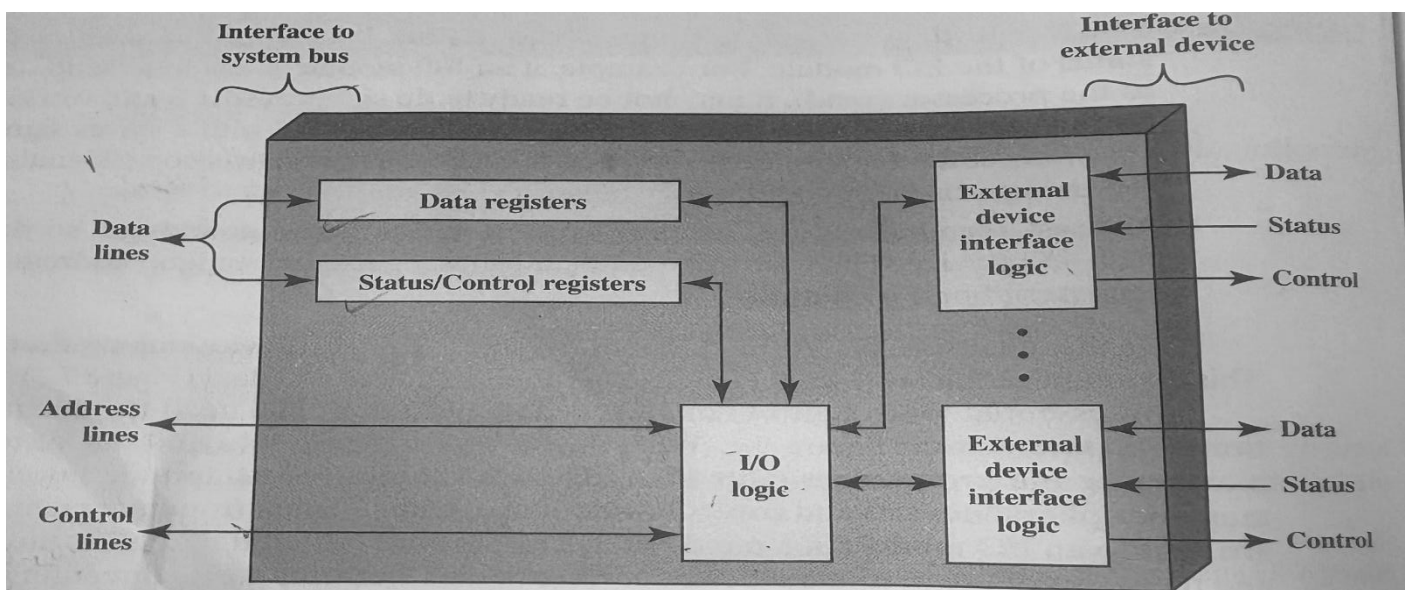


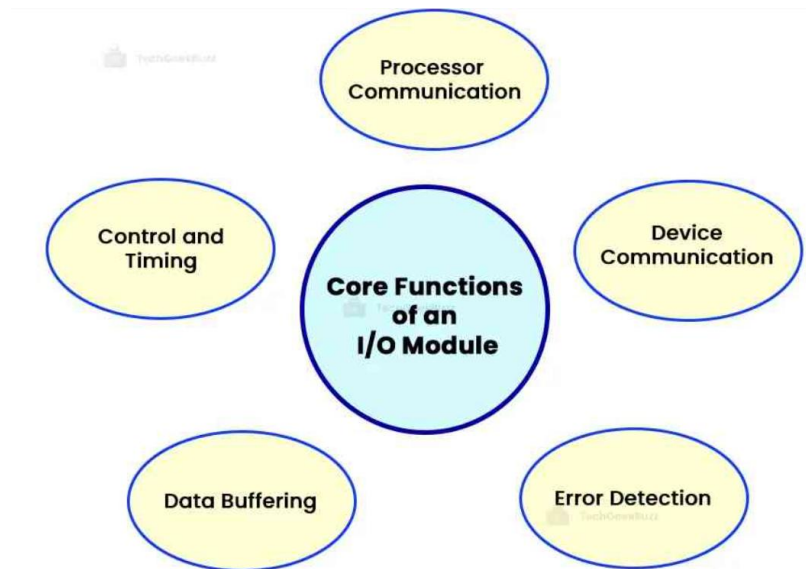
Figure 7.3 Block Diagram of an I/O Module

The I/O module connects the rest of the computers through the set of signal lines (**Data, address and control lines**). Data transferred to and from CPU are buffered at *data registers*. *Status register* provides current status information and control register accept control information from CPU. *I/O logic* interacts with CPU through address and control lines. Control lines are used to issue commands to I/O e. The module must also recognize the unique address associated with devices it controls.

I/O module functions to allow the processor to view a wide range of devices. It hides the details of an external device so that a processor can function in terms of simple *read* and *write* commands.

Core Functions of an I/O Module

The following are the major functions of an I/O module:



1. Processor Communication: When the I/O module communicates with the processor, it involves the following

- **Command Decoding:** The I/O module accepts and decodes commands from the processor.
- **Data Exchange:** It facilitates data transfer between the processor, memory, and peripheral devices.
- **Status Reporting:** The I/O module communicates the status of each peripheral device to the CPU, as they are slower.
- **Address Decoding:** It organizes each peripheral device by recognizing its unique address.

2. Control and Timing: Another major function of the I/O module is control and timing to coordinate between internal resources and external devices. Here is how the data transfers from external devices to the processor:

- The processor asks the i/o module about the status of the attached peripheral device.
- The I/O module returns the status of a specific device.
- If the device is ready to transmit the data, the processor issues a command to the I/O module to transfer data.
- The I/O module then takes the requested data from the external device and transmits it to the processor.

3. Data Buffering: Data buffering helps the I/O module to manage the data transfer speed for the data sent by the processor to peripheral devices. This helps peripheral devices and the processor work at the same pace.

4. Error Detection: The I/O module detects electrical and mechanical malfunctions in peripheral devices and reports them to the processor. The parity bit method is one of the ways the I/O module detects errors.

5. Device Communication: It also facilitates data exchange between different peripheral devices.

Why Need an I/O Module?

The foremost reason to use an I/O module is to establish a connection between the processor and I/O devices.

- A computer's processor is an electronic device, whereas I/O devices are electronic, mechanical, and electrical. As a result, the I/O module standardizes the signals from different I/O devices and transmits them to the processor.
- The data transfer rates of the processor and memory are faster than I/O devices. Being an intermediate device, the I/O module facilitates communication between the processor or memory and slow I/O devices. It translates all faster instructions from the processor and sends them to I/O devices in a format they can understand and process.
- A computer connects with a variety of peripheral devices. Hence, the processor becomes overloaded with the logic of multiple peripherals. It creates obstacles to new developments. The I/O module can solve this problem, too.